

Gary Hobson

MAT260 Module 5

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8.5-2

Given: The initial AES key is 128 bits of all 1s. So:

$$W(0) = W(1) = W(2) = W(3) = \text{FF FF FF FF}$$

(1) Compute $W(4)$ to $W(7)$:

$$W(4) = W(0) \oplus T(W(3))$$

Using the AES key schedule, this results in:

$$W(4) = \text{E8 E9 E9 E9}$$

$$W(5) = W(1) \oplus W(4) = \text{17 16 16 16}$$

$$W(6) = W(2) \oplus W(5) = \text{E8 E9 E9 E9}$$

$$W(7) = W(3) \oplus W(6) = \text{17 16 16 16}$$

So:

$$W(4) = W(6) = \text{E8 E9 E9 E9}$$

$$W(5) = W(7) = \text{17 16 16 16}$$

Also:

$$W(5) = \sim W(4) \quad (\text{bitwise complement})$$

(2) Show $W(10) = W(8)$ and $W(11) = W(9)$:

From the key schedule:

$$W(8) = W(4) \oplus W(7)$$

$$W(9) = W(5) \oplus W(8)$$

$$W(10) = W(6) \oplus W(9)$$

$$W(11) = W(7) \oplus W(10)$$

Since:

$$W(5) \oplus W(6) = \text{FF FF FF FF},$$

and XORing the same value twice cancels it out:

$$W(10) = W(8)$$

$$W(11) = W(9)$$

Confirmed.

8.5-6

If my machine can test 2^{56} keys per second, but the AES keyspace has 2^{128} keys.

To find how long it will take:

$$\frac{2^{128}}{2^{56}} = 2^{72} \text{ seconds}$$

Convert seconds into years:

$$2^{72} \approx 4.7 \times 10^{21} \text{ seconds}$$

There are approximately 3.15×10^7 seconds in a year:

$$\frac{4.7 \times 10^{21}}{3.15 \times 10^7} \approx 1.49 \times 10^{14} \text{ years}$$

So, it would take approximately 149 trillion years for the machine to brute force all 2^{128} AES keys even with a speed of 2^{56} keys per second. This confirms the astronomical strength of AES-128 against brute force attacks.